A Generative AI Framework for Inferring Consistent Ocean Statistics at Depth Using Sea Surface Data

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ABSTRACT: Understanding subsurface ocean dynamics is essential for quantifying oceanic heat and mass transport, but direct observations at depth remain sparse due to logistical and technological constraints. In contrast, satellite missions provide rich surface datasets—such as sea surface height, temperature, and salinity—that offer indirect but potentially powerful constraints on the ocean interior. Here, we present a probabilistic framework based on score-based diffusion models to reconstruct three-dimensional subsurface velocity and buoyancy fields, including the energetic ocean eddy field, from surface observations. Using a 15-level primitive equation simulation of an idealized double-gyre system, we evaluate the skill of the model in inferring the mean circulation and the mesoscale variability at depth under varying levels of surface information. We find that the generative model successfully recovers key dynamical structures and provides physically meaningful uncertainty estimates, with predictive skill diminishing systematically as the surface resolution decreases or the inference depth increases. These results demonstrate the potential of generative approaches for ocean state estimation and uncertainty quantification, particularly in regimes where traditional deterministic methods are underconstrained or ill posed.

KEYWORDS: Ocean; Transport; Turbulence; Inverse methods; Neural networks; Uncertainty

1. Introduction

Inferring subsurface ocean dynamics from surface observations remains a fundamental challenge in physical oceanography. The ocean interior plays a crucial role in regulating Earth's climate by transporting heat, carbon, and other tracers, yet direct measurements at depth are sparse due to logistical and technological limitations. In contrast, satellite missions have provided decades of continuous, high-resolution observations of sea surface height (SSH) and temperature and large-scale salinity observations, offering a vastly richer dataset than what is available below the surface (Liu and Hu 2024). Theoretical and numerical studies suggest that surface anomalies—such as sea surface height—strongly correlate with subsurface structures at mesoscale and submesoscale scales, meaning surface data could provide meaningful constraints on the ocean interior (Hurlburt 1986; Cooper and Haines 1996). However, existing methods for reconstructing subsurface states from surface observations are often limited by strong dynamical assumptions or high computational costs.

This study focuses on reconstructing interior velocity and temperature fields, with a particular emphasis on their correlations, which are essential for estimating mass and heat transport throughout the ocean. Mesoscale eddies, which account for 20%–50% of total heat transport (Su et al. 2018; Saenko et al. 2018), are particularly difficult to infer from surface observations alone. A deterministic reconstruction, which attempts to match a single "true" state, may fail to capture these dynamically significant features. Instead, we take a probabilistic approach, where a generative model reconstructs a distribution of plausible subsurface states statistically consistent with available surface data. This methodology ensures that eddy statistics, such as their variance, spatial correlations, and contributions to heat transport, are preserved without erasing dynamically important details.

Several physically motivated methods have been developed to infer subsurface ocean states from surface data, each with distinct advantages and limitations. The surface quasigeostrophic (SQG) framework has been used for subsurface econstruction. The theory works well when the inherent assumptions in QG theory are satisfied and there exists information about the vertical stratification (Held et al. 1995; Lapeyre and Klein 2006; Wang et al. 2013; Liu et al. 2019). Theoretical extensions, such as the SQG model with a multivariate empirical orthogonal function, have been shown to improve upon standard SQG by accurately reconstructing both surface- and subsurface-intensified eddies, particularly in regions with complex vertical stratification like the Kuroshio Extension and the Peru–Chile upwelling system (Yan et al. 2020). However, recent satellite missions such as the Surface Water and Ocean Topography (SWOT) mission (Fu et al. 2024) achieve higher resolution than traditional altimeters (down to 10 km) and resolve scales where the QG approximations are no longer valid, necessitating additional theoretical development or alternative approaches.

Theoretical limitations can be overcome when abundant data are available. In particular, machine learning offers a data-driven alternative to augment or completely replace physically constrained methods for ocean state estimation. Its applications range from model emulation (Dheeshjith et al. 2024) to data

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assimilation (Champenois and Sapsis 2024), with increasing success in leveraging satellite observations. Linear models have demonstrated the statistical predictability of large-scale ocean currents, such as the Atlantic meridional overturning circulation (AMOC), using satellite-derived data (Cromwell et al. 2007; DelSole and Nedza 2022). More recently, feed-forward neural networks (FNNs) and convolutional neural networks (CNNs) have been applied to reconstruct meridional overturning circulations from ocean bottom pressure, zonal wind stress, and sea surface properties (Solodoch et al. 2023; Meng et al. 2024). Additionally, studies incorporating satellite and Argo float data have demonstrated skill in inferring velocity at depth (Chapman and Charantonis 2017), while CNNs trained on quasigeostrophic (QG) simulations have successfully mapped surface streamfunction anomalies to subsurface flow fields in idealized settings (Bolton and Zanna 2018). Complementary efforts using QG theory (George et al. 2021) and data assimilation techniques (Martin et al. 2023) have refined our understanding of large-scale ocean dynamics. Meanwhile, operationally constrained models integrate observational data with numerical simulations, leading to more accurate reconstructions of subsurface ocean conditions (Solodoch et al. 2023).

While these studies have demonstrated that satellite-derived surface information can be used to infer the ocean interior, most have relied on supervised learning, where a model is trained to minimize residuals against a single true state. However, a given surface ocean state does not deterministically correspond to a unique deep ocean state. A more physically sound approach requires probabilistic modeling, which allows for generating an ensemble of plausible subsurface states constrained by surface observations, while preserving the uncertainties and variability inherent in ocean dynamics. Probabilistic methods can be considered an extension of supervised learning methods as they can often reproduce the inference from such methods, as a subset of their capabilities. For example, minimizing the root-mean-square error of a probabilistic output results in the mean of the underlying distribution which is the target of supervised learning methods (barring discrepancies in training and architecture choices).

Importantly, probabilistic modeling allows for uncertainty quantification (UQ) (Forget et al. 2015; Souza et al. 2020): Not only can samples from the inferred distribution be explored but these samples also provide estimates of plausible subsurface fields when surface information is insufficient to constrain the output fully. Generative modeling offers a flexible approach to sampling from such distributions, with score-based diffusion models emerging as a particularly effective tool for high-dimensional statistical inference. Generative-adversarial models (GANs) have also been applied to predict ocean temperature profiles from sea surface temperature (Meng et al. 2023; Zhang et al. 2023), but these works did not leverage the potential for uncertainty quantification. The study in Martin et al. (2025) leveraged score-based generative methods in the context of ocean surface data assimilation, but again, they did not focus on uncertainty quantification; however, there have been recent efforts at examining uncertainty quantification in the presence of sparse ocean-like measurements in a quasigeostrophic model (Babu et al. 2025). Score-based generative methods in particular have also been explored when addressing extreme event sampling (Stamatelopoulos and Sapsis 2025). Beyond conditional sampling, score-based generative models have demonstrated utility in diverse applications, including calculating response functions (Giorgini et al. 2024, 2025c), estimating the dimensionality of data manifolds (Stanczuk et al. 2023), stabilizing the dynamical evolution neural network models (Pedersen et al. 2025), parameter calibration (Giorgini et al. 2025b), data assimilation (Manshausen et al. 2025), or devising stochastic analogs to chaotic equations (Giorgini et al. 2026; Giorgini 2026; Giorgini et al. 2025a).

This study develops a probabilistic generative approach to infer the ocean interior state conditional on observable surface variables, such as sea surface height. By framing subsurface inference as a conditional sampling problem, we aim to quantify the variability and uncertainty of ocean interior states while leveraging the wealth of information encoded in surface observations. Rather than simply predicting a single deterministic state of ocean currents from the surface, our objective is to determine to what extent observations of the ocean surface constrain the ocean interior state while ensuring that the whole subsurface ocean state, including correlations among different variables, are properly captured. As we will show, a well-trained model can even predict the extent to which it is wrong in its predictions.

With these aims in mind, the rest of the paper is organized as follows. In section 2, we provide the physical context for the study, introduce the numerical simulation that we use as a testbed for the generative method, and describe the generative modeling framework. Results are presented in section 3, where we apply the generative model to the ocean simulation and assess its predictive skill in reconstructing subsurface velocities and temperature alongside their variability as a function of sea surface height. We also repeat the exercise with spatially filtered sea surface heights to understand how less informative surface conditions alter predictions. Section 4 concludes and discusses further extensions of this work.

2. Numerical simulations and generative AI model

In the following subsections, we describe the details of the ocean model output, the generative artificial intelligence (AI) model, and training. In essence, we will use an ocean simulation to generate a time series of the ocean state, consisting of the SSH, velocities, and temperature at various vertical levels. This dataset then forms the backbone for training a neural network to determine plausible velocities and temperatures given a sea surface height.

a. Ocean simulation

We illustrate the generative AI method with an idealized simulation rather than reanalysis data or a more realistic ocean setup for two primary reasons: 1) With an idealized study, it is possible to generate (regenerate) as much training data as needed to test the methodology and 2) a more realistic ocean model introduces additional challenges like measurement uncertainty, data gaps, and nonstationary statistics. We

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believe that one must first demonstrate that the proposed methodology works in an idealized setting here before attempting it in a more complex scenario.

As with any idealized simulation, we focus on particularly salient features that generalize to more complex scenarios. Thus, the role of the simulation is not to capture realistic ocean features but rather to serve as a testing ground for whether statistical inference from the free surface to the ocean interior is possible. With this in mind, we choose a baroclinic double gyre, similar to Cox and Bryan (1984), as a representative ocean simulation for the methodology. The simulation generates circulation patterns consistent with those observed in the subtropical and subpolar gyres. We will not run the ocean model to full statistical equilibrium so that we can study both statistically stationary and nonstationary features (Hasselmann 1976). We use a strict interpretation of "full statistically stationary" to imply that "all" metrics of the simulation are converged. In our simulation, only some statistics achieve stationarity during spinup, while others do not. The generative AI approach implicitly assumes stationarity of all statistics if it is to extrapolate future states from past ones. We will, therefore, note when the inference problem is applied to stationary versus nonstationary statistics.

Using Oceananigans.jl (Ramadhan et al. 2020; Silvestri et al. 2024; Wagner et al. 2025a; Silvestri et al. 2025; Wagner et al. 2025b), we solve the Boussinesq equations under the hydrostatic approximation. The dynamical core uses a nonlinear free surface and $z^\star$ coordinates. The prognostic variables are the horizontal velocities U and V, the SSH η, and the temperature T. A linear equation of state relates the buoyancy of seawater B to the temperature (Roquet et al. 2015). The figures will refer to the Oceananigans simulation output as "OcS" for brevity.

The idealized ocean domain extends from 15° to 75°N in latitude and from 0° to 60° in longitude with walls on all sides. The domain is 1800 m deep with a flat bottom. A zonal wind stress is applied at the surface, independent of longitude, with a cosine profile peaking eastward (positive) in the middle of the domain at a value of 0.1 N m^{-2} and at a minimum and westward (i.e., negative) at the northern and southern walls. The surface temperature is restored to a linear profile in latitude ranging from 30°C on the southern boundary at 15°N to 0°C at the northern boundary at 75°N. A quadratic bottom drag is imposed at the seafloor, and convective adjustment is used for vertical mixing. The (dimensionless) quadratic drag coefficient is 10^{-3}, and we use a horizontal diffusivity and viscosity of 2.5×10^2 m^2 s^{-1}, a background vertical diffusivity of 10^{-5} m^2 s^{-1}, and background vertical viscosity of 10^{-2} m^2 s^{-1}.

The domain is discretized into $256 \times 256 \times 15$ grid points, leading to a 25-km resolution in latitude. The longitudinal resolution depends on the given latitude and is calculated by multiplying the latitudinal resolution with the cosine of the latitude, which ranges from factors of 0.96 to 0.26. The vertical grid spacing varies from 50 m at the surface to 180 m at depth. The simulation is initialized with zero velocity, a flat sea surface, and uniform buoyancy and runs for ~425 years. The ocean circulation spins up in the first ~100 years, which are discarded in the following analysis. Once spun up, the solution is characterized by a double-gyre circulation with convection along the northern boundary and a rich mesoscale eddy field. However, the ocean depths have not yet reached statistical equilibrium and continue to evolve.

We show the resulting fields and their evolution in time in Fig. 1. From left to right, each column shows the SSH η, the zonal U, meridional V, vertical velocity W, and temperature T at a 400-m depth. Each row is a snapshot at years 20, 200, and 300. There is a cyclonic gyre in the northern half of the domain and an anticyclonic gyre in the lower half. The convection in the north half of the domain cools fluid parcels that sink and fill the abyss with cold water. The resulting simulation exhibits a nontrivial depth-dependent structure, deviating from equivalent barotropic behavior, and includes a substantial ageostrophic component at all vertical levels.

The goal of this manuscript is to infer plausible ocean states (U, V, W, T) at all depths from the instantaneous SSH field η. We use this simulation as a proxy for performing ocean inference given satellite information and a model for the ocean interior. We only focus on the SSH since the imposed forcing constrains the present simulation's surface temperature field. In a more realistic setup, we would use more information.

Our simulation is not in statistical equilibrium for all observables of the flow, as can be observed from Fig. 2, which shows the L^1 norm (the average absolute value) of all variables at a few selected depths, e.g.,

$$\|V\|_1 = \frac{1}{60^\circ}\frac{1}{60^\circ}\int_{15^\circ}^{75^\circ}\int_0^{60^\circ} |V(\lambda, \varphi, z, t)| d\lambda d\phi. \tag{1}$$

We use this measure as it is sensitive to localized fluctuations, such as the root-mean-square value. Including metric terms does not change the overall trend and is unnecessary for the present case, but we will include them later when calculating physically meaningful quantities rather than statistical measures of nonstationarity. The initial spinup period (blue) is discarded. Thus, all the data in the work use the time periods associated with the middle portion (red) and the end portion (orange) of the simulation. Closer to the surface (first column), it appears that a quasi-statistical equilibrium has been reached, whereas at depth (last column), statistical equilibrium has not been reached, especially for temperature. This quasi-equilibrium nature of the ocean is a feature we wished to capture in our dataset when investigating the limits of statistical inference predicated on past data. We comment that the sea surface height's L^1 norm is also not in equilibrium. Stated differently, the model is trained on the "data distribution" (red) but evaluated on the later "validation distribution" (orange), so this distribution shift is itself an additional source of error. That being said, there are many statistics which can, for the most part, be regarded as being in statistical equilibrium such as the surface horizontal velocity components.

b. Probabilistic models

In the present work, we draw samples from subsurface fields, U, V, W, and T, given the sea surface height η, where the underlying probability distribution comes from the data distribution generated by the time series of the primitive

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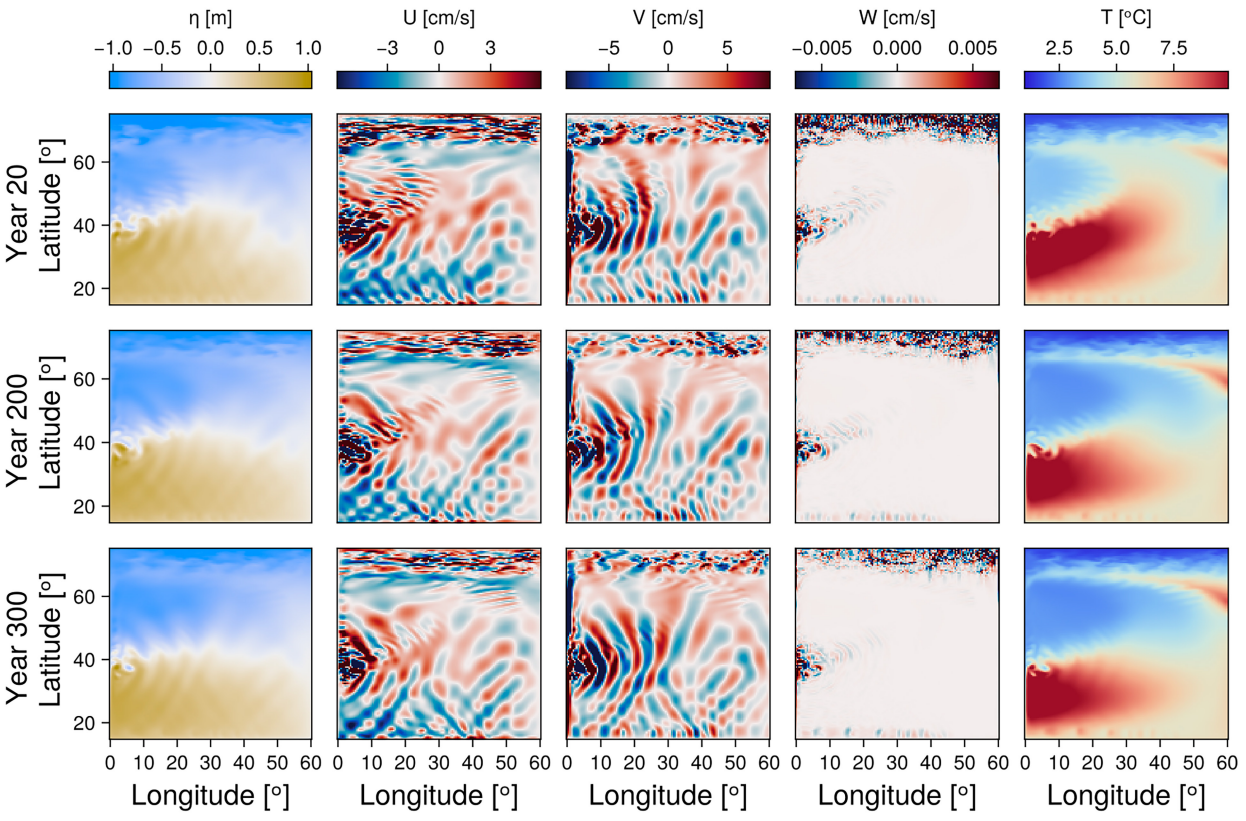


FIG. 1. Snapshots of the ocean state from a double-gyre simulation. Each column is a different ocean state, and each row is a fixed moment in time. (top) Year 20, (middle) year 200, and (bottom) year 300. (first column) The SSH, (second column) the zonal velocity U, (third column) the meridional velocity V, (fourth column) the vertical velocity w, and (fifth column) the temperature T. The velocities and temperature are evaluated at a 400-m depth.

equations. Mathematically, this is represented as a conditional probability distribution, $\mathcal{P}$:

$$\text{conditional distribution} = \mathcal{P}(\text{subsurface fields}|\text{surface fields}). \tag{2}$$

The conditional probability distribution outputs four fields U, V, W, and T at every grid point in three-dimensional space, conditioned on the SSH η. Concretely, samples from this distribution produce $4 \times 256 \times 256 \times 15$ numbers conditioned on 256×256 numbers using the full resolution of the simulation. This high-dimensional conditional probability distribution is learned from the dataset using deep learning methods, see section 2c.

If the conditional information is perfectly informative, there is a functional relationship, $\mathcal{F}$, between SSH and predicted fields. In this case, the conditional distribution collapses to a delta function:

$$\mathcal{P}(\text{subsurface fields}|\text{surface fields}) = \delta(\text{subsurface fields} - \mathcal{F}[\text{surface fields}]). \tag{3}$$

If the provided information is "not useful" or maximally uninformative, then the conditional probability distribution, $\mathcal{P}$, should reduce down to the marginal distribution of the data, e.g., a random sample with statistics similar to those the generative model has been trained on independently of the surface information:

$$\mathcal{P}(\text{subsurface fields}|\text{surface fields}) = \mathcal{P}_M(\text{subsurface fields}), \tag{4}$$

where $\mathcal{P}_M$ is the marginal distribution of the subsurface fields. Most inference tasks lie between these two extremes, Eqs. (3) and (4). We comment that we use the word "marginal" as is commonly used in the statistics literature with regard to distributions.

Traditional machine learning methods find a functional form $\mathcal{F}$ between input and outputs. This methodology is well suited for problems where such a functional form exists. Otherwise, traditional methods that minimize a root-mean-square (RMS) error to find a functional form can only infer the conditional expectation associated with Eq. (2), that is,

$$\mathcal{F}[\text{surface fields}] = \mathbb{E}[\text{subsurface fields}|\text{surface fields}], \tag{5}$$

much like a "mean" statistic minimizes the RMS error of a random variable (see the appendix for a proof of this result). If the conditional distribution is a delta function, as in Eq. (3),

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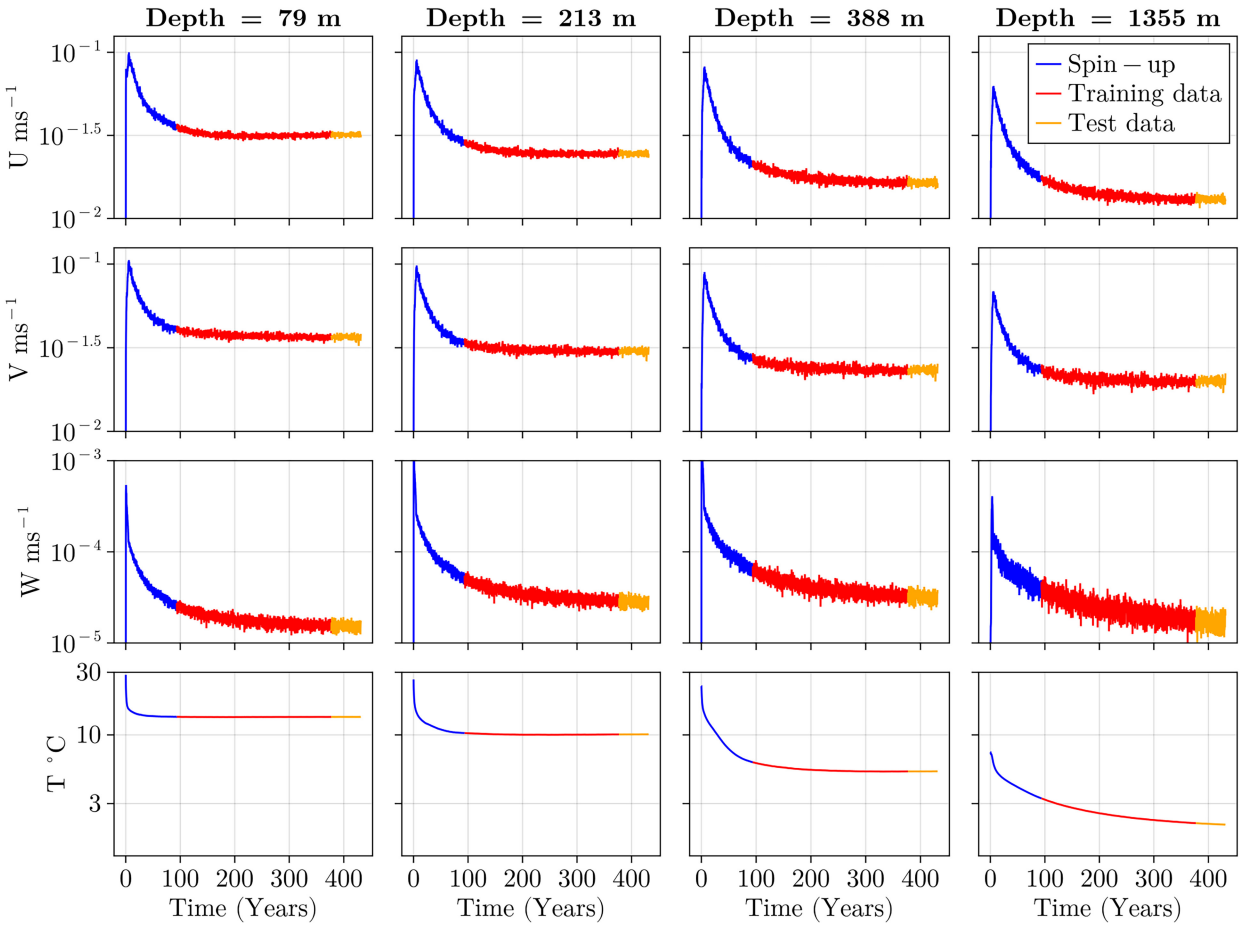


FIG. 2. Evolution of L^1 norms in time. We show the average value of the absolute value of various fields [see Eq. (1)] as a function of time in years on a logarithmic scale. The surface fields are in quasi-statistical equilibrium, whereas the fields near the bottom of the domain are still converging, especially for the temperature field. The different colors represent the discarded spinup data (blue), training data (red), and test data (orange) for training a score-based diffusion model.

little is lost by performing the average; however, if there is significant variance in the output field, then the best that a deterministic model can do is to average over such variance. At the extreme, the output of a deterministic model will produce the ensemble average of the previously seen data, e.g., the average value of the marginal distribution, Eq. (4). We surmise that this observation partially explains why fields drawn from a conditional distribution better represent the statistical properties of fluid flows compared with those learned via a deterministic functional relationship. Details will not be "washed away" from averaging.

An important consequence of Eq. (5) is that we can compare generative models to traditional regression techniques indirectly by taking the ensemble average of the output given by the generative model. In this way, we can see what details are missed by using nongenerative approaches as well as provide a baseline for when we expect traditional approaches to outperform generative methods. This observation is independent of the neural network architecture and comes instead directly from the objective function that is being minimized by an "RMS loss" of a traditional method versus the "denoising score-matching loss" of a generative approach.

c. *Score-based diffusion model*

Our work leverages recent advancements in score-based generative modeling, particularly building upon foundational contributions (Song and Ermon 2019, 2020; Ho et al. 2020; Karras et al. 2022). These methods allow for efficient sampling of arbitrary probability distributions, including conditional distributions.

The key deep learning methodologies to sample the conditional distribution include reverse diffusion and conditional denoising score matching (Anderson 1982; Hyvärinen and Dayan 2005; Vincent 2011; Batzolis et al. 2021). Architecturally, we adopt the U-Net framework (Ronneberger et al. 2015) and a well-established model utilized in generative modeling (Deck and Bischoff 2023; Bischoff and Deck 2024).

The simulated SSH data are coarse grained from its original resolution of $256 \times 256 \times 15$ to $128 \times 128 \times 15$ by averaging neighboring horizontal points to expedite the training of the

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generative model. This lower-resolution dataset loses few features from the original but dramatically expedites the training process since the entire dataset can be loaded into computer memory. We take monthly snapshots over the simulation period to ensure that our data are sufficiently decorrelated.

A coarse-grained sea surface height is the conditional information provided to the model, which serves as the primary input for generating samples. We examine various SSH coarse-graining levels to determine the subsurface fields' predictability. Specifically, we average neighboring points by factors of 2 until only a single value for the SSH remains as input for the deep learning model. That is to say, we progress from a 50-km resolution to a 6400-km resolution in latitude. This procedure yields eight possible SSH inputs at each time, one for each resolution level. Each state, U, V, W, T, and η, is separately normalized at each of the 15 levels of depth so that each field has the spatiotemporal mean removed and is divided by twice the spatiotemporal standard deviation. This rescaling keeps the fields roughly $\mathcal{O}(1)$ for training purposes. Thus, 8 numbers are used separately to rescale U, V, W, and T at each of the 15 vertical levels, and 2 numbers are used to rescale η, leading to a total of $8 \times 15 + 2$ scaling factors.

We train a single neural network that simultaneously outputs the entire 3D field for U, V, W, and T and takes in all coarse-grained inputs. We enable the latter step by randomly choosing among the 8 different coarse-grained SSH. We use a 2D U-Net with 61 input channels, 60 output channels (15×4 2D output fields and 1 2D input field), and $\mathcal{O}(10^7)$ parameters. The resulting network has a $\mathcal{O}(10^2)$ megabytes memory footprint, which we compare to the $\mathcal{O}(10^2)$ gigabytes used to train the generative model. In this sense, we can think of the generative model as a statistical compression technique which, in the present case, reduces the storage by a factor of 10^3. The Adam optimizer (Kingma and Ba 2014) is employed for training the network.

3. Results

Given the ocean simulation data (OcS) from section 2a and the neural network architecture from section 2c, we perform statistical inference by drawing 100 sample fields using the reverse diffusion process, given SSH. We test the neural network by feeding SSH from the last 50 years of the double-gyre simulation which were not used during training. The predictions of the subsurface variables are tested for the various coarse-grained SSH input data.

The generative model draws many sample realizations from a single SSH field, yet we only have one ground-truth simulation to compare against for a given SSH. Accordingly, we compare the distributional output to the ground truth in two ways. The first is to compare the ensemble mean of the generated samples to the ground truth, since the ensemble mean is relatively easy to compute and serves as a good "average" guess. This choice is most useful when the conditional information is sufficiently informative, as in the limit described by Eq. (3). The second is to compare the ocean simulation output to the full set of ensemble members, which yields a probability distribution. We can then compare this distribution to one obtained by taking one generated sample (e.g., the first) and comparing it to the others (e.g., 2–100). We then assess whether the two probability distributions are consistent. This latter choice is a purely "data-driven" way to determine how accurate the model's predictions are. A well-trained model (which requires sufficient data and an appropriate choice of architecture) makes meaningful estimates of the range of plausible fields for a given SSH, and thus, the two resulting distributions should be similar. We will return to this point later.

a. *Baseline performance of the generative model*

We start by examining how well the SSH constrains the three components of velocity and temperature at a 400-m depth. We show the result of a trained neural network model for generating conditional samples in Fig. 3. The sea surface height (first column) and corresponding simulated variables at a 400-m depth (second column) are taken from 50 years in the future from the end of the training set. Each row corresponds to a different ocean state component (U, V, W, T), as labeled. Using the generative model, we draw 100 samples from the conditional distribution with this sea surface height as the conditional information. We then take the ensemble average over these samples (third column). We show 2 of the 100 samples in the fourth and fifth columns to illustrate the "generative" nature of the prediction from the neural network.

There is overall agreement between the ground truth (Fig. 3, second column) and the expected value of the model predictions (Fig. 3, third column). This result means that a substantial ageostrophic component is also captured. The ensemble mean of the generated samples is also similar to the ground truth and the representative generated samples, consistent with the limited sample diversity. The conditional distribution is nearly a delta function, suggesting that the sea surface height is very informative of the ocean state and that a functional relationship can be found between the sea surface height and subsurface currents. In other words, the present inference task is leaning toward Eq. (3). Given that a traditional supervised learning task would produce the ensemble mean prediction of a generative model, low sample diversity would imply that a supervised learning task would perform well for the horizontal velocity components and the temperature field; however, the vertical velocity field has significant sample diversity, particularly in the northern region, implying that a traditional machine learning model would miss these fluctuations and "average them out."

The sixth column in Fig. 3 shows the absolute value of the difference between the OcS and the AI mean, and the seventh column shows the ensemble standard deviation of the AI method. We represent predictive uncertainty by using the generated fields' standard deviation at each grid point. The largest standard deviation is associated with the northern convecting region, especially for vertical velocity. This observation suggests that the sea surface height alone is insufficient to constrain the pointwise vertical velocity magnitude in these regions. A traditional machine learning model would average out these fluctuations to produce the average vertical velocity field as given by the "AI Mean" column. We see the flexibility

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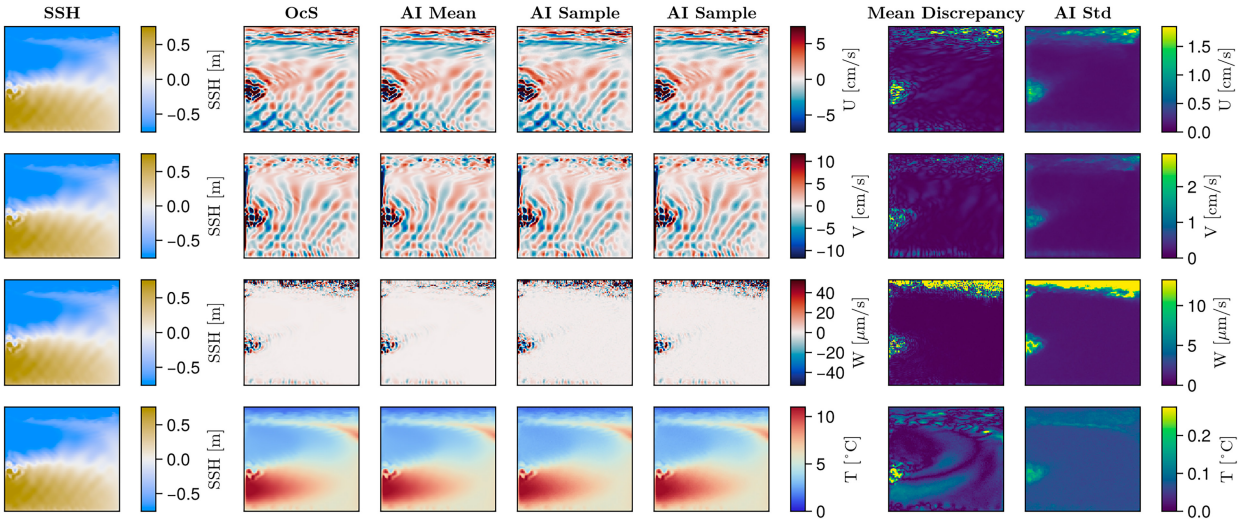


FIG. 3. Statistical inference of the ocean interior state from SSH. Given the (first column) SSH input field, we reconstruct the (second column) ocean state: the velocity component fields U, V, W, and temperature T. All fields are generated simultaneously. We show (third column) the mean over 100 samples predicted by the neural network and (fourth column),(fifth column) two representative samples. (sixth column) The absolute difference between the AI mean and the OcS ground truth alongside (seventh column) the AI standard deviation.

of the generative model in informing our ability to apply traditional machine learning models. The second largest source of uncertainty is the eddying gyre region, which contributes to the uncertainty of the velocity components. The mean discrepancy column and the AI Std column have similar spatial structures, suggesting that the method can identify regions in which it is less capable of prediction.

Although mean discrepancy and the standard deviations are similar in structure and magnitude across the velocity fields, we see that the temperature field has a different spatial structure. Although the strongly eddying region and the northern convective remain the largest source of uncertainty, there is also a uniform background uncertainty in the domain as well. This uniform uncertainty of the temperature field could perhaps be related to the nonstationarity of the domain average temperature field, which is a statistic that evolves on a much longer time scale than others in the OcS. Ultimately, the relative errors of the temperature field are smaller as compared to the velocity fields. The standard deviation of the generative model's temperature field overestimates this (small) uncertainty.

b. *Ocean state predictions and uncertainties with degraded surface height input*

Next, we assess how well the model predicts the ocean interior as sea surface height information is progressively removed. We do this by applying a filter to the sea surface height, averaging neighboring points, and providing this filtered free surface as input to the generative model. We use the same neural network model for this inference task because it was trained simultaneously on all coarse-grained fields. We show the resulting statistical inferences for the zonal velocity in Fig. 4 and quantitative errors as a function of depth and coarse graining in Fig. 6.

We first focus in Fig. 4. In the first column, the top row shows SSH at 200-km latitudinal resolution (32×32 points), the second row shows 800-km resolution (8×8 points), and each subsequent row applies an additional factor-of-2 coarsening, yielding 1600- and 3200-km resolutions, respectively. Although the network predicts all the velocity components and buoyancy at each vertical level, we restrict the visualization to the zonal velocity at a 400-m depth. The outputs can be directly compared to the first row in Fig. 3. The generative model produces 100 samples for each coarse-graining level. The second column shows the ensemble average zonal velocity, and columns three–five show randomly selected samples. We see by inspection that, as we proceed downward through the rows, the ensemble average departs progressively from the samples and the sample diversity increases. In addition, we show the pointwise standard deviation of the 100-member ensembles in the last column to quantify the diversity of the generated samples.

These observations suggest that an SSH coarse grained beyond $\mathcal{O}(1000)$ km is no longer informative since the generated samples become so diverse as to reproduce statistics of the data distribution, a "worst case" scenario for the generative method. As we coarse grain the free surface (lose information), the inference task moves from something informative [similar to Eq. (3)] toward the uninformative [similar to Eq. (4)] marginal distribution of the training data. Taken together, we see that the generative model can quantify the "nonexistence" of a unique purely deterministic method for predicting subsurface states; the problem becomes inherently ill posed from a deterministic perspective (i.e., trying to predict a single unique subsurface state is not possible because the given information is insufficient to constrain potential subsurface states). For a deterministic method to succeed, it must

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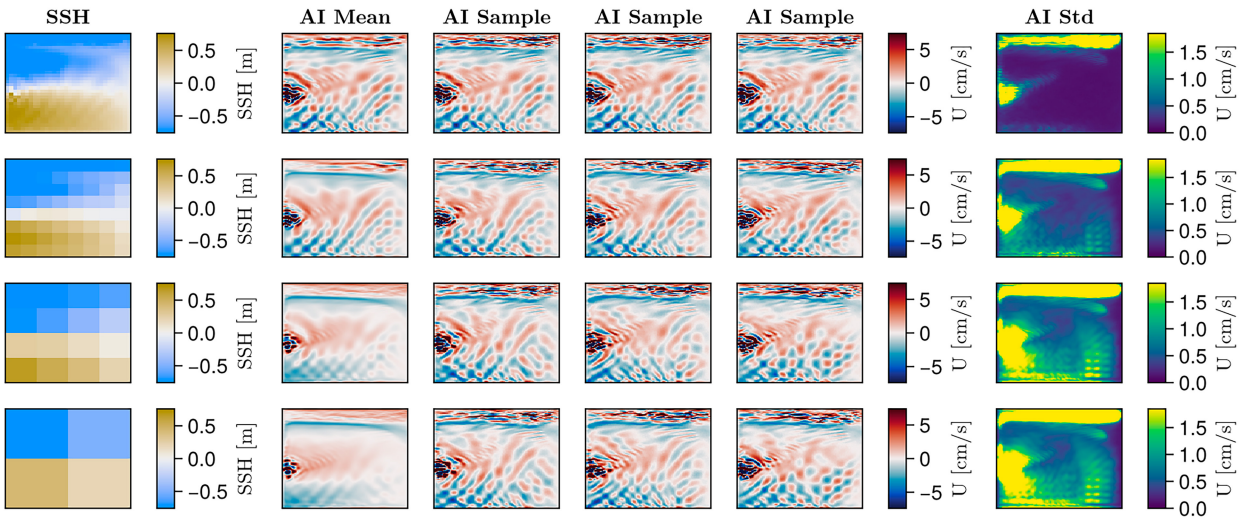


FIG. 4. Statistical inference of zonal velocity given the different resolutions of the free surface. Given (first column) the SSH at several coarse-grained resolutions, we show the output of the generative AI model ensemble in the subsequent columns. We show the average value over 100 samples predicted by (second column) the neural network and (third column)–(fifth column) representative samples. (sixth column) The pointwise standard deviation of the generative AI ensemble. The latitudinal resolutions of SSH are, from top to bottom, 200, 800, 1600, and 3200 km.

be supplemented with either additional information, some regularization, or a criterion to select from the multitude of possible subsurface configurations (e.g., choosing the most probable subsurface state, an ensemble mean subsurface state, or some other selection criteria).

We compare the sampled distribution using the most uninformative (i.e., fully coarse grained over the entire domain) SSH as the conditional information to the marginal distribution of the data over the training period in Fig. 5. This corresponds to the maximally uninformative case, and the goal is to show that, when the input is uninformative, the model reverts to drawing samples with statistics similar to those of the training data. To do so, we compare the ensemble statistics of the generative model to the temporal statistics of the training data. For brevity, we use a fixed depth. The rows correspond to the different fields U, V, W, and T at a 400-m depth. The OcS mean/std is a temporal mean/std over the training dataset, and the AI mean/std is generated from a 100-member ensemble using only the domain average free-surface height as input, i.e., we remove spatial structure from the input field. We see that there are still wave-like oscillations present in the AI predictions, which we attribute to insufficient averaging from the limited generated samples, since using fewer ensemble members exacerbated the oscillations; overall, however, there seems to be good agreement between the two calculations.

In Fig. 6, we quantify the discrepancy between the ensemble average of the AI prediction and the ocean simulation in predicting the various fields as a function of resolution and depth. We take the difference between the OcS simulation and each ensemble member individually. We use the L^1 norm (average absolute error) to illustrate the discrepancy and the resulting distribution with different bands whose contours represent the 0.6, 0.7, 0.8, and 0.9 quantiles, from darker to lighter. The points are the ensemble average of the resulting distribution. We denote this as the "OcS AI Discrepancy." We also perform an analogous computation purely from the AI ensemble, which we call the "AI Shuffle Discrepancy" in the second row. Here, we plot the distribution given by the difference between different ensemble members. Specifically, we take the L^1 norm between ensemble members 1 and 2, 2 and 3, 3 and 4, and so forth to ensemble members 100 and 1, yielding 100 samples of an L^1 norm, which we then take to be a distribution to compare against the distribution given by the OcS AI Discrepancy. A similar calculation is to take ensemble member 1 as a "fake" ground truth and take the difference between the other ensemble members 2–100. We choose the "shuffled" version to reduce variation due to a "unlucky" choice of ground truth ensemble member.

We see that (for all fields), as we decrease the resolution of the sea surface height, the discrepancy between generated fields and the OcS increases. The average discrepancy of the U and V velocity field components decreases as a function of depth since their magnitudes decrease with depth. The temperature field T has the opposite trend, with the largest average discrepancies coming from the surface, perhaps due to the larger overall temperature values near the ocean's surface. The vertical velocity exhibits a less clear trend, with the largest average discrepancy coming from a 400-m depth and the lowest coming from the near-surface value. A potential explanation lies in the fact that $w = 0$ at the surface and the seafloor, so the largest value should come from somewhere at middepth.

The trends of the first row are quantitatively reflected in the second row. The "Shuffle Discrepancy" is similar to the difference between the simulation output and the AI ensemble. This observation suggests that the AI method can provide

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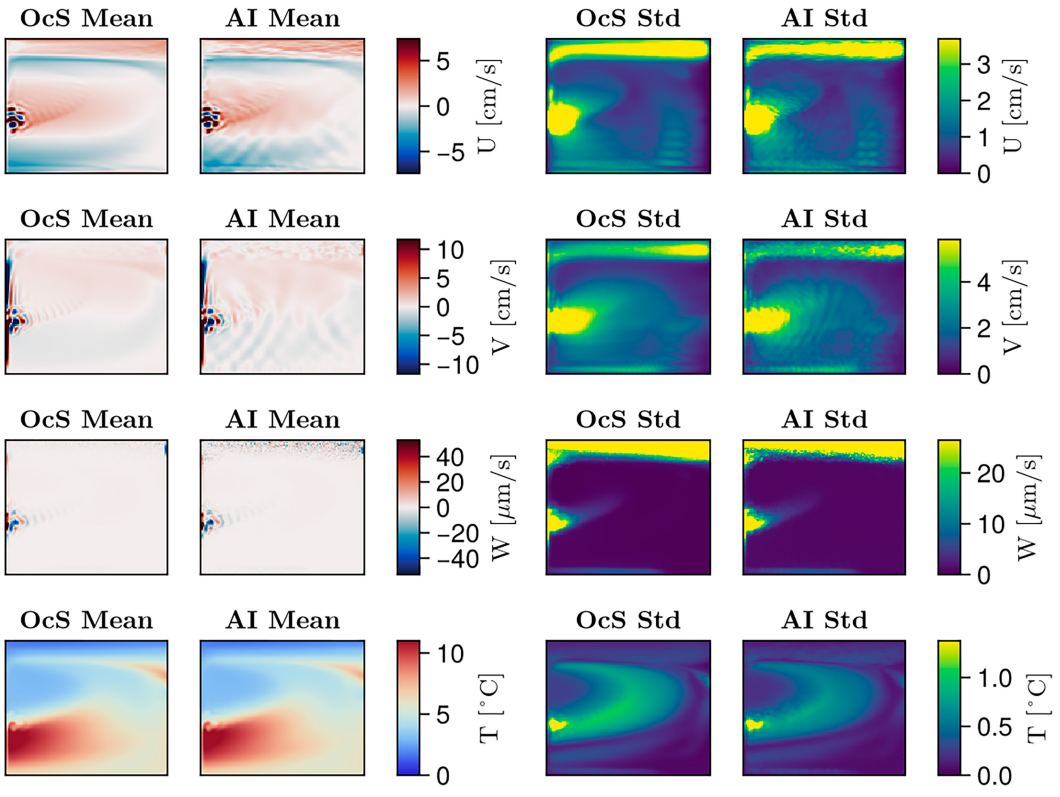


FIG. 5. Comparison of the AI ensemble mean to the OcS temporal mean. Using fully coarse-grained SSH as input to the network, we generate 100 samples and calculate pointwise statistics. We compare these ensemble statistics to the temporal analog of the ocean simulation over the training period.

meaningful uncertainty estimates in so far as the data it was trained on are similar in distribution to what are seen in the future. For example, we see a difference between the 1355-m depth "OcS AI" temperature discrepancy and the corresponding AI Shuffle Discrepancy where the AI Shuffle Discrepancy underpredicts the associated error. We attribute this to the lack of equilibrium of the ocean simulation depths, as reflected in Fig. 2.

c. *Heat flux predictions and uncertainties with degraded surface height input*

Thus far, we have only examined statistics within a given field but have not looked at correlations across fields. We address this by calculating the vertical and meridional eddy heat flux. We do not show the full heat flux because its value depends on arbitrary temperature units. Also, we use the zonal averages of the fluxes for illustrative purposes. Mathematically, we use

$$\bar{\cdot} \overset{\text{def}}{=} \frac{1}{60^\circ}\int_{0^\circ}^{60^\circ} \cdot \, d\lambda, \tag{6}$$

$$\text{vertical eddy heat flux} = \cos(\varphi)(\overline{wT} - \overline{w}\,\overline{T}), \tag{7}$$

$$\text{meridional eddy heat flux} = \cos(\varphi)(\overline{vT} - \overline{v}\,\overline{T}). \tag{8}$$

We show the heat fluxes as a function of latitude at various depths (in each column) in Fig. 7 for the highest-resolution SSH. The "ground truth" results are the blue lines, and the generative AI results with 100 samples are shown as ribbon plots that capture the 0.6, 0.7, 0.8, and 0.9 quantiles from darker to lighter shadings of red. The top row is the vertical heat flux, and the bottom row is the meridional heat flux, both as a function of latitude. Generally, we see that the generative model predicts the heat flux well for all cases, with the ribbons increasing in relative size with depth: The meridional heat flux at the lowest shown depth is not as well constrained in relative terms.

We also calculate the eddy heat fluxes as a function of SSH resolution at a 400-m depth in Fig. 8. Similar to the previous plot, the top row is the vertical heat flux; the bottom row is the meridional heat flux; the ground truth results are the blue lines; and the generative AI results with 100 samples are shown as a ribbon with 0.6, 0.7, 0.8, and 0.9 quantiles from darker to lighter shadings of red. The heat flux prediction grows in uncertainty as we coarse-grain the sea surface height. The loss of information here is understood upon examining the first column in Fig. 4. As the SSH field is increasingly coarse grained, we lose precise information about the location of the jet separation point—the region where the western boundary current detaches from the wall and begins to meander eastward. This separation point is critical in organizing mesoscale eddy activity and lateral heat transport. When spatial detail in the SSH is removed, the model can no longer

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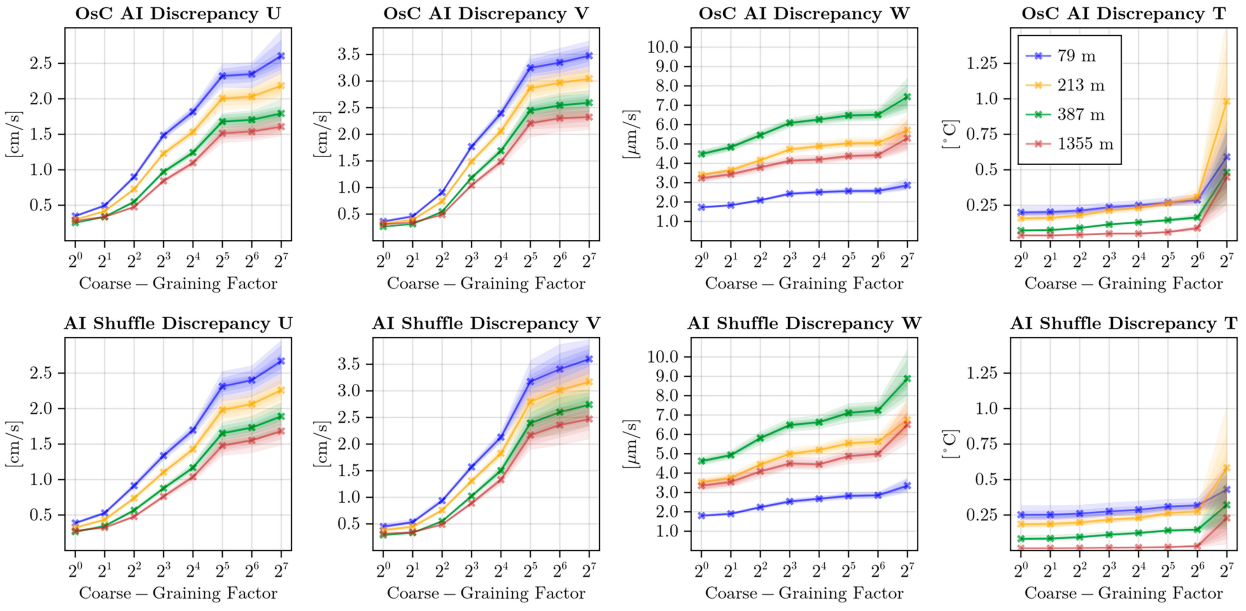


FIG. 6. Discrepancy as a function of depth and coarse graining for a fixed SSH. (top) We show the distribution of the L^1 discrepancy between the 100-member neural network ensemble and the ocean simulation. (bottom) We show an analogous AI-only discrepancy computed from L^1 differences between ensemble members. The columns represent the different fields, U, V, W, and T. Within a panel, the different colors represent different depths, and the horizontal axis corresponds to different coarse-graining factors of SSH with 2^0 representing the full resolution (50 km) and 2^7 representing an average value of SSH (6400-km resolution).

resolve where the jet begins to meander, leading to increased uncertainty in the predicted eddy heat fluxes, particularly in the downstream region where the current interacts with ambient stratification. The ability of the generative model to produce plausible eddy fluxes in the ill-posed regime is another feature of generative models: A deterministic model would be relegated to making an approximation of the form $\mathbb{E}[vT] \approx \mathbb{E}[v]\mathbb{E}[T]$, where the expected value on the right-hand

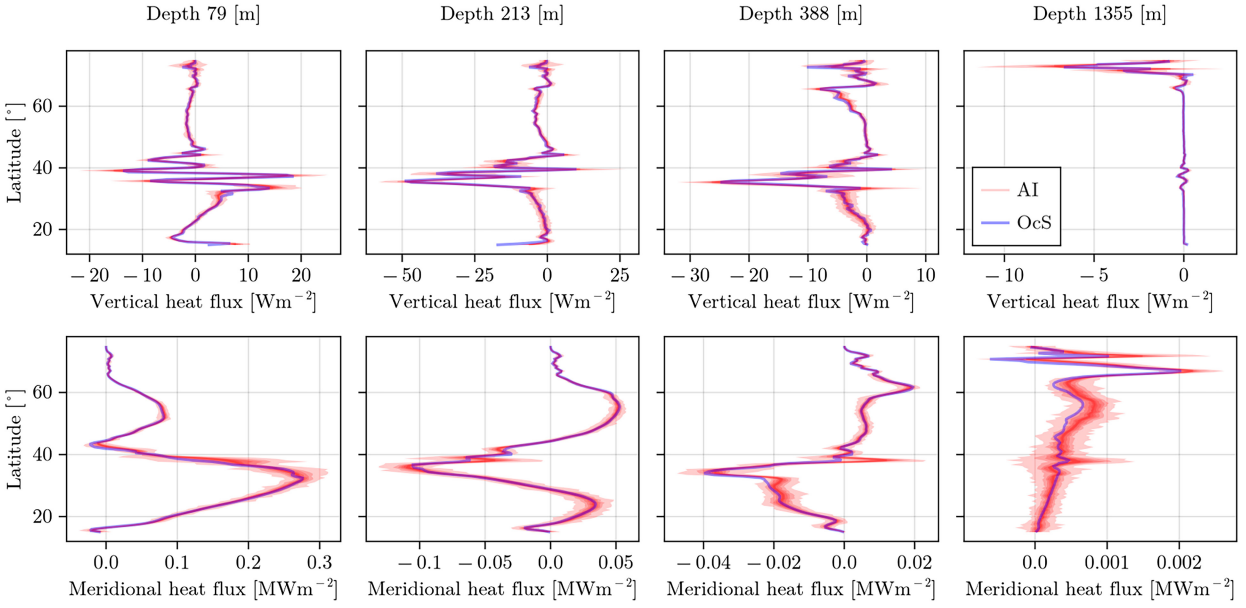


FIG. 7. Zonal average of vertical and horizontal eddy heat flux for a fixed SSH at various depths. We show the zonal average of the (top) vertical eddy heat flux and (bottom) meridional eddy heat flux as functions of latitude at a fixed time and several depths in blue. The neural network results using the full-resolution SSH as input are shown in red, with the solid red line representing the ensemble average heat flux. The ribbons represent the 0.6, 0.7, 0.8, and 0.9 quantiles, from darker to lighter, as computed with the 100-member ensemble. Each column shows a different depth, and the relative uncertainty (red ribbon region) increases as we go deeper into the domain.

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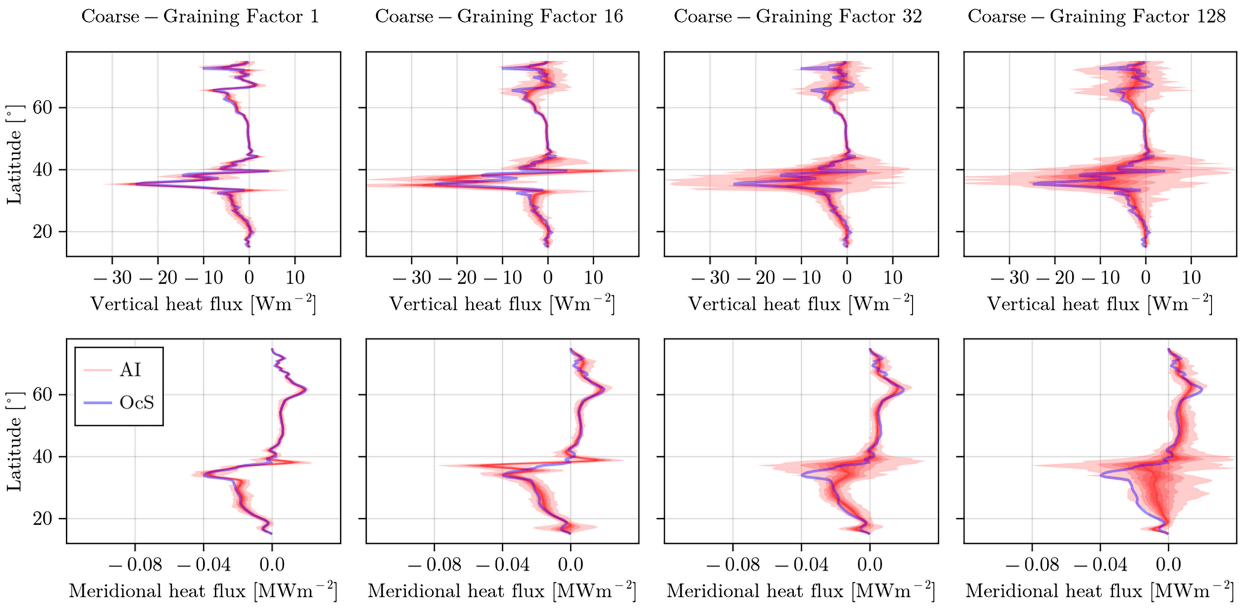


FIG. 8. Zonal average of vertical and horizontal eddy heat flux for a fixed SSH at a 400-m depth. We show the zonal average of the (top) vertical eddy heat flux and (bottom) meridional eddy heat flux over the first 15° of the domain, as functions of latitude at a fixed time and a depth of 400 m, in blue. The neural network results are shown in red, with the solid red line representing the ensemble average heat flux. The ribbons represent the 0.6, 0.7, 0.8, and 0.9 quantiles, from darker to lighter. Each column uses a different level of coarse-grained SSH. As the amount of coarse graining increases, the confidence bounds broaden and approach the marginal distribution of the training data. Information about the location of the meander is lost when the resolution is reduced from a factor of 8 to a factor of 16. See the first column and third row of Fig. 4 for the corresponding SSH.

side is the output of the deterministic model's prediction for v and T. This occurs for the same reason mentioned prior that a traditional machine learning model can (at best) produce the ensemble mean of what a generative model can produce.

d. *Statistics combining various depths*

Understanding how well generative models predict subsurface ocean dynamics at different depths is crucial for assessing their viability in state estimation and climate modeling. Next, we assess the generative model's ability to aggregate statistics across different vertical levels. Specifically, to evaluate the model's performance, we compare the predictions of the generative model against the ocean simulation for correlations across depths, the zonally and vertically integrated meridional heat flux, and the meridional overturning streamfunction in depth coordinates. We do this for the same 50-yr future SSH used in previous sections.

Figure 9 quantifies the vertical correlation of the subsurface fields under various inference settings. Here, we use the correlation coefficient $\mathcal{C}$. This quantity for the meridional velocities at two different depths V_1 and V_2 is calculated as

$$\mathcal{C}(V_1, V_2) = \frac{\mathbb{E}[(V_1 - \mathbb{E}V)_1(V_2 - \mathbb{E}V_2)]}{\sqrt{\mathbb{E}[(V_1 - \mathbb{E}V_1)^2]}\sqrt{\mathbb{E}[(V_2 - \mathbb{E}V_2)^2}}, \tag{9}$$

where the expected value is over the latitude–longitude points (i.e., a horizontal average without the metric terms), at a fixed depth, and for a fixed ensemble member (when appropriate). A similar calculation is performed for other quantities. For ensemble-based comparisons, results are shown as ribbon plots, with shading corresponding to the 0.6, 0.7, 0.8, and 0.9 quantiles (from dark to light).

The left column in Fig. 9 shows the correlation between each field and its surface value, computed from the OcS simulation (top) and the generative AI model (bottom). These panels confirm that the generative model reproduces the vertical coherence present in the simulation. Furthermore, we note that the simulation deviates from equivalent barotropic behavior due to the changing correlation coefficient. The middle column compares AI-generated fields to the OcS profiles across depth, conditioned on either full spatial-resolution SSH (top) or mean (i.e., no spatial structure) SSH (bottom). The stronger correlations in the top row indicate that the fine-scale structure in SSH provides a meaningful constraint on subsurface variability. The existence of some level of correlation in the lower row reflects the similarity between past ocean states and future ocean states as given by the simulation. After all the fields are not "arbitrary" but rather related to the past flow states of the model, e.g., regions in the past with no flow will remain with no flow. The right column presents ensemble-based correlations among AI samples conditioned on identical SSH inputs (top: full resolution; bottom: no resolution), serving as a "shuffle test" analogous to the discrepancy analysis in Fig. 6. These correlations largely mirror the patterns seen in the middle column, indicating internal consistency in the generative model's outputs. A notable

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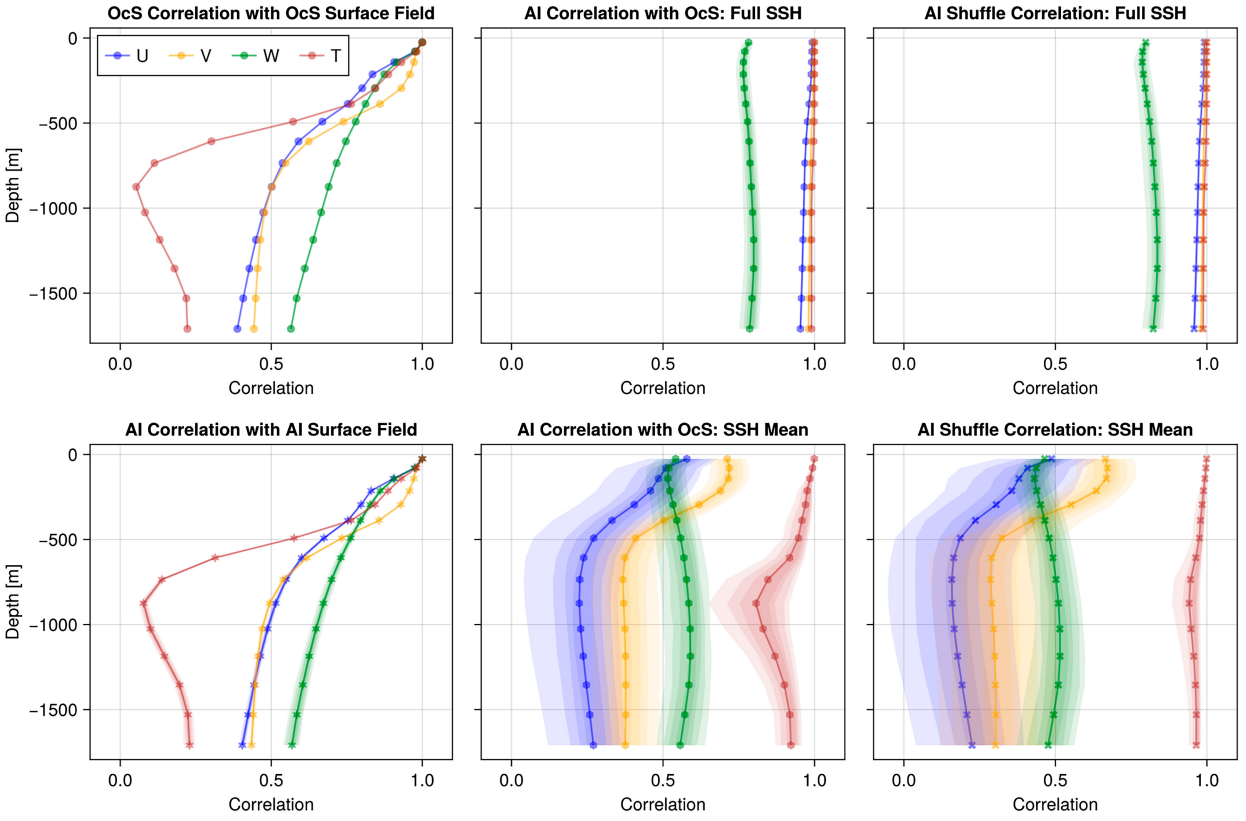


FIG. 9. Correlation coefficients. (left) Correlations with the surface field in (top) OcS and (bottom) the generative AI outputs. (middle) Cross correlations between AI-predicted fields and OcS profiles, using (top) full-resolution SSH and (bottom) domain-averaged SSH as conditioning inputs. (right) Ensemble-based correlations computed purely from the AI model for (top) full-resolution SSH and (bottom) domain-averaged SSH.

exception is seen in the temperature field at depth, where correlations drop significantly—a result we attribute to the lack of statistical equilibrium in the abyssal ocean.

We show in Fig. 10 the generative AI's ability to reproduce the zonally and vertically integrated meridional heat flux:

$$\text{meridional heat flux} = \rho_0 c_p R \cos(\varphi) \int_{-1800\text{m}}^{0\text{m}} \int_{0^\circ}^{60^\circ} V(\lambda, \varphi, z, t) \times T(\lambda, \varphi, z, t) d\lambda dz, \tag{10}$$

where R is the radius of Earth, c_p is the heat capacity of water, and ρ_0 is the reference density. The red line denotes the ensemble mean of the generative AI prediction, and the shaded ribbons show the 0.6, 0.7, 0.8, and 0.9 quantiles from dark to light. These bands correctly encapsulate the ground truth ocean simulation prediction. This quantity is evaluated 50 years in the future, similar to the previous sections. The uncertainty is quite large, yet the ensemble mean prediction is similar to ground truth. The uncertainty can likely be reduced by improving the model by better accounting for the vertical structure. Furthermore, the generative method need not be nondivergent, which may affect the resulting statistics.

Last, we calculate the meridional overturning streamfunction:

$$\psi(\varphi, z, t) \stackrel{\text{def}}{=} R \cos(\varphi) \int_{z}^{0} \int_{0^\circ}^{60^\circ} V(\lambda, \varphi, z', t) d\lambda dz'. \tag{11}$$

Figure 11 shows the resulting overturning streamfunction. The top left panel shows the ocean simulation output, the top

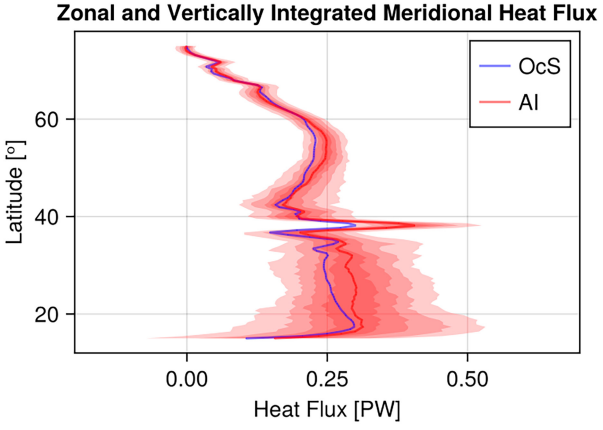


FIG. 10. Zonal and vertically averaged heat transport. The instantaneous transport as a function of latitude is estimated from the model simulation (blue line) and with the AI model using only SSH information (red line). The pink shading represents the 0.6, 0.7, 0.8, and 0.9 quantiles, from darker to lighter.

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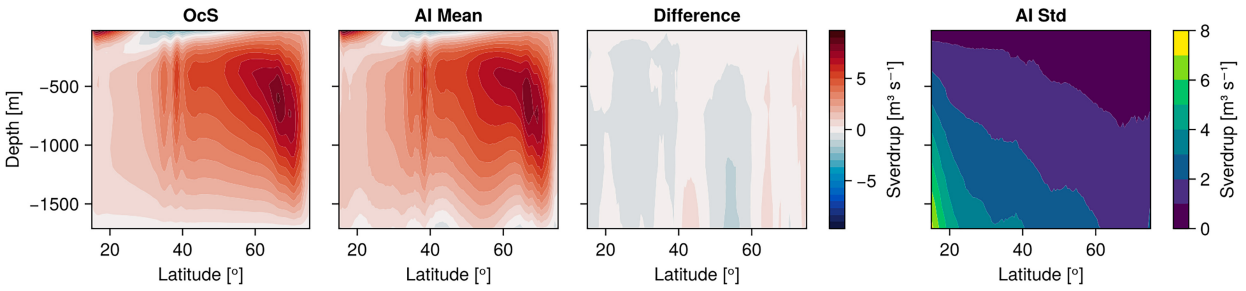


FIG. 11. The instantaneous overturning streamfunction in depth coordinates. Here, we show the overturning streamfunction at each vertical level z and latitude for a fixed moment in time. (first column) The ocean simulation is compared to (second column) the AI mean. (third column) The difference between the ocean simulation and the AI mean and (fourth column) the AI standard deviation.

middle panel shows the mean of the 100 generated samples, and the top right panel shows the difference. The bottom left and bottom middle panels show arbitrarily chosen samples, and the bottom right panel shows the AI uncertainty, measured as the standard deviation of the 100 generated samples.

We see that the overall structure in the upper 400 m of the domain is well captured, with errors increasing with depth. Furthermore, we see a larger discrepancy at lower latitudes and middepth. The largest standard deviation of the AI ensemble is at low latitudes and at depth. The lack of statistical equilibrium contributes to the error here. The largest error comes from the pool of cold water slowly finding its way along the seafloor, from northern latitudes toward the lower latitudes. This process takes time and was not shown to the neural network during training. Regardless, the resulting variability is overestimated.

4. Conclusions

This study has demonstrated the potential of generative modeling as a statistical framework for inferring the interior state of the ocean from surface observations. We explored the ability of diffusion-based generative models to reconstruct subsurface velocity and buoyancy field statistics using only information about the free surface in a statistically nonequilibrated system, offering a probabilistic approach to state estimation. Our key findings can be summarized as follows:

1) We have demonstrated, in an idealized setting, that ocean state estimation can be formulated as a conditional generative modeling problem, where score-based diffusion models sample from the distribution of subsurface velocity and buoyancy fields given surface data. This approach captures inherent uncertainties in subsurface reconstructions, providing plausible subsurface fields consistent with the given information.
2) The generative AI method accurately reconstructs key ocean circulation features—including mesoscale eddies, zonal jets, eddy-heat flux, and overturning circulations—across different depths, dynamical regimes, and surface data resolutions. Notably, the generative approach provided meaningful uncertainty estimates, particularly in eddying and convective regions.
3) While the model performs well in regions where surface observations strongly constrain subsurface dynamics, its predictive skill diminishes at greater depths and in regions where the ocean state differs substantially from the states seen during training.

Our results demonstrate that generative models might capture the ocean interior's statistical variability, even when conditioned on coarse-resolution surface data. However, decreasing surface observations' spatial resolution or increasing inference depth diminishes predictive skill. In the limiting case, when the surface signal is insufficient to constrain the interior, the generative model reverts to sampling from the marginal distribution of the training data. As shown in Fig. 6, the most pronounced deterioration in performance occurs when the surface resolution is degraded from 100 km (coarsening factor 2^2) to 200 km (2^3), particularly for the horizontal velocity components where the eddy length scale is on the order of 100 km. For temperature, predictive accuracy declines when large-scale gradient information is lost. These findings suggest that improvements in satellite resolution—from current AVISO-class missions (200 km) to SWOT (50 km)—could significantly enhance our ability to infer subsurface ocean dynamics (Ballarotta et al. 2019; Fu et al. 2024). Incorporating additional dynamical priors, such as quasi-geostrophic constraints or hybrid physics-informed machine learning architectures, may further improve the fidelity and robustness of generative reconstructions.

Future work will extend this approach to more complex ocean simulations that include seasonal cycles, bathymetry, and finer-scale turbulence, providing additional constraints on the generative model. Furthermore, integrating observational datasets with multiple modalities such as sea surface salinity, Argo float profiles, sea surface temperature, and velocity measurements from moorings will enable direct evaluation of the model's performance in real-World Ocean conditions (Martin et al. 2025). Sparse measurements can also be incorporated as a part of the generative framework (Babu et al. 2025). To properly incorporate satellite altimetry would be a space–time version of the "in-painting" exercise of that work, leading to a 4D probabilistic reconstruction.

Measurement uncertainty must also be accounted for. One avenue for doing so would be to draw from a distribution of consistent free-surface heights, effectively sampling

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from $\mathcal{P}$(subsurface|surface)$\mathcal{P}_M$(surface), where $\mathcal{P}_M$(surface) incorporates uncertainty in the surface field. Incorporating theoretical predictions as additional "input" for the generative model will also be explored. This way, we can use a theory as a "physics-informed" guess, further refined through the generative method and a numerical ocean simulation with higher fidelity physics. For example, one could use theory to provide further conditional information to augment the prediction of the generative method. Symbolically, we would be drawing from the distribution $\mathcal{P}$(subsurface fields|surface fields, theory), which could possibly further reduce ensemble spread in samples.

The methods developed in this study contribute to the broader effort of integrating machine learning with physical modeling in climate science. By leveraging advances in generative AI and ocean state estimation, this work provides a foundation for a new approach to data assimilation, parameterization, and statistical inference in ocean modeling. Such methodologies can also be applied to atmospheric geophysical models as well as Earth system models to provide a probabilistic view of multiscale dynamics.

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Data availability statement. Scripts for reproducing the figures and the neural network training are located at https://github.com/sandreza/DoubleGyreInference. The Oceananigans script for producing simulations is located at https://github.com/CliMA/Oceananigans.jl/blob/as/z-star-coordinate/double_gyre_5.jl. We developed JAX code implementing the deep learning method at https://github.com/sandreza/JaxDiffusion.

APPENDIX

A Proof of Conditional Expectation

In the main text, it was claimed that the minimizer of the mean-square-error (MSE) is the expectation of a conditional distribution. Here, we provide a standard proof of this claim for completeness.

Minimizing the MSE of a dataset is equivalent to minimizing the cost functional $\mathcal{C}$,

$$\mathcal{C}[\mathcal{F}] = \int \mathcal{P}_J(\text{subsurface fields, surface fields}) \| \mathcal{F}(\text{surface fields}) - \text{subsurface fields} \|^2, \tag{A1}$$

across all possible functions $\mathcal{F}$. Here, $\mathcal{P}_J$ is the "data distribution," i.e., the empirical distribution that constitutes the data pairs (surface fields, subsurface fields), as given by the numerical simulation. For brevity, we will use "x = surface fields" and "y = subsurface fields" and x^n to denote the surface field at time step t_n and y^n to denote the subsurface field at time step t_n. We denote the total number of time steps in the dataset as N. With these abbreviations, we have

$$\mathcal{P}_J(x, y) = \frac{1}{N} \sum_{n=1}^{N} \delta(x - x^n)\delta(y - y^n), \tag{A2}$$

as our explicit data distribution, where δ is the Dirac delta function; however, a well-trained neural network will regularize the above data distribution so that it may generalize to unseen data pairs. Ideally, the regularized distribution should be related to the "true distribution" of the underlying numerical simulation.

Our cost functional with the same notation is

$$\mathcal{C}[\mathcal{F}] = \int \mathcal{P}_J(x, y) \| \mathcal{F}(x) - y \|^2. \tag{A3}$$

To minimize the above functional across all functions $\mathcal{F}$, we take the variational derivative of the above expression with a small function perturbation $\delta\mathcal{F}(x)$ and neglect higher-order terms to yield

$$\delta\mathcal{C} = \mathcal{C}[\mathcal{F} + \delta\mathcal{F}] - \mathcal{C}[\mathcal{F}] \approx 2 \int \mathcal{P}_J(x, y)\delta\mathcal{F}(x) \times [\mathcal{F}(x) - y]. \tag{A4}$$

We now use the identity $\mathcal{P}_J(x, y) = \mathcal{P}_M(x)\mathcal{P}(y|x)$ to rewrite the above expression as

$$\int \mathcal{P}_J(x, y)\delta\mathcal{F}(x)\cdot[\mathcal{F}(x) - y] = \int dx \mathcal{P}_M(x)\delta\mathcal{F}(x) \times \int dy \mathcal{P}(y|x)[\mathcal{F}(x) - y]. \tag{A5}$$

Carrying out the integral over y, we identify

$$\int dy \mathcal{P}(y|x)\mathcal{F}(x) = \mathcal{F}(x) \quad \text{and} \quad \int dy \mathcal{P}(y|x) y = \mathbb{E}[y|x], \tag{A6}$$

which reduces our functional derivative to

$$\delta\mathcal{C} = 2 \int dx \mathcal{P}_M(x)\delta\mathcal{F}(x) \times [\mathcal{F}(x) - \mathbb{E}[y|x]]. \tag{A7}$$

Setting $\delta\mathcal{C} = 0$ and noting that this must hold across all variations δF yields the expression

$$\mathcal{F}(x) = \mathbb{E}[y|x], \tag{A8}$$

i.e., the minimizer is the conditional expectation.

At this point, it is worth contrasting the above expression with what is obtained by a generative method. The generative model directly tries to capture (and sample from) the conditional distribution $\mathcal{P}(y|x)$, whereas the deterministic method calculates directly the expected value $\mathcal{F}(x) = \mathbb{E}[y|x]$. Thus, to compare generative models to traditional models

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(minimizing with respect to MSE), we generate samples from the distribution $\mathcal{P}(y|x)$ and take an empirical average afterward.

There are, however, practical issues that may prevent such an identification of traditional machine learning methods with generative methods. One issue comes from different architectures or forms of implicit regularization that are used when training neural networks. Another issue is that one may require a large number of empirical samples to generate a particular statistic. If either model overfits on the training dataset, then these results do not apply since the connection between the two only happens for the "global" minimizer of the cost functional for the underlying data distribution (as opposed to the empirical data distribution).

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